

Chuanruo Ning

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EDUCATION

Cornell University

Ph.D. in Computer Science. Advisors: Wei-Chiu Ma, Kuan Fang

Ithaca, NY, United States

Aug. 2024 – Present

Peking University

Bachelor of Science in Computer Science

Beijing, China

Aug. 2020 – Jun. 2024

RESEARCH EXPERIENCE

Open-world manipulation by equipping VLM with interactive digital twins

Sep. 2024 – May. 2025

Advisors: Prof. Kuan Fang, Prof. Wei-Chiu Ma

Cornell University

- Build physically grounded interactive digital twins that can provide action-conditioned outcome generation.
- Prompt VLM with the future states for model predictive control.
- Enables diverse open-world manipulation which requires semantic and physical understanding.

3D Part Recognition from 2D Image with a Single 3D Annotation

Jun. 2023 – Jan. 2024

Advisor: Prof. Alan Yuille

Johns Hopkins University

- Explore zero-shot object part segmentation that only requires a single 3D annotation for part definition.
- Establish 3D to 3D correspondence for uniform category-level 3D representation.
- Establish 3D to 2D correspondence for render-and-compare based part detection.

Few-shot Affordance Learning for Articulated Objects

Dec. 2022 – Jul. 2023

Advisors: Prof. Hao Dong, Dr. Kaichun Mo

Peking University

- Efficiently manipulate unseen articulated objects in novel categories with minimal explorations.
- Propose to measure semantic similarity between local geometries across different categories.
- Enable model to perform few-shot learning on novel categories by discovering uncertain yet important areas.

Foresightful Deformable Object Manipulation

Jan. 2022 – May. 2023

Advisor: Prof. Hao Dong

Peking University

- Learn dense visual representations that reveal the dynamic and kinematic properties of deformable objects.
- Propose a novel training pipeline to take the future potential of object states into consideration.
- Train the model in a reversed step-by-step manner to make it foresightful.

PUBLICATIONS AND MANUSCRIPTS

* indicates equal contributions or advising

- Maximus Pace*, Prithwish Dan*, **Chuanruo Ning**, Atiksh Bhardwaj, Wei-Chiu Ma, Kushal Kedia. X-Diffusion: Training Diffusion Policies on Cross-Embodiment Human Demonstrations. Under Review, 2025.
- **Chuanruo Ning**, Kuan Fang*, Wei-Chiu Ma*. Prompting with the Future: Open-World Model Predictive Control with Interactive Digital Twins. **In RSS, 2025.**
- **Chuanruo Ning***, Jiawei Peng*, Yaoyao Liu, Jiahao Wang, Yining Sun, Alan Yuille, Adam Kortylewski, Angtian Wang. Part321: Recognizing Object Parts in 3D from a 2D Image Using 1-Shot Annotations. Under Review, 2025.
- Yitong Li*, Ruihai Wu*, Haoran Lu, **Chuanruo Ning**, Yan Shen, Guanqi Zhan, Hao Dong. Broadcasting Support Relations Recursively from Local Dynamics for Object Retrieval in Clutters. **In RSS, 2024.**
- **Chuanruo Ning**, Ruihai Wu, Haoran Lu, Kaichun Mo, Hao Dong. Where2Explore: Few-shot Affordance Learning for Unseen Novel Categories of Articulated Objects. **In NeurIPS, 2023.**
- Kai Cheng*, Ruihai Wu*, Yan Shen, **Chuanruo Ning**, Guanqi Zhan, Hao Dong. Learning Environment-Aware Affordance for 3D Articulated Object Manipulation under Occlusion. **In NeurIPS, 2023.**
- Ruihai Wu*, **Chuanruo Ning***, Hao Dong. Learning Foresightful Dense Visual Affordance for Deformable Object Manipulation. **In ICCV, 2023.**

SELECTED HONORS AND AWARDS

- Huatai Securities Scholarship, Peking University, 2024.
- Merit Student, Peking University, 2023.
- John Hopcroft Scholarship, Peking University, 2022.
- Dean's Scholarship, Peking University, 2022.

ACADEMIC SERVICE

- Conference Reviewer: CVPR 2024-2026, NeurIPS 2024-2026, AAAI 2024-2025, ICLR 2025-2026, ICRA 2026, 3DV 2026.